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Inventor(s)	Morrison; Jimmy et al.

AIR BLASTER TOY

Abstract

In various aspects, an air cannon includes a handle having a handle body and a ring at a proximal end of the handle body. The air cannon may also include a bag extending between a first end and a second end, the first end of the bag securable to an interior of the ring. The air cannon may further include a coil disposed about a circumference of the bag, with the coil extending from a first end to a second end. In some embodiments, the first end of the coil is engaged by the interior of the ring to secure the first end of the bag to the interior of the ring. The air cannon may include a collapsed state and an expanded state.

Inventors: **Morrison; Jimmy (Springville, UT), Thompson; Nathan (Twin Falls, ID)**

Applicant: **Frustration Free Solutions, LLC** (Twin Falls, ID)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] A claim for priority to the Feb. 29, 2024 filing date of U.S. Provisional Patent Application No. 63/559,266, titled AIR BLASTER TOY (the '266 Provisional Application "), is hereby made pursuant to 35 U.S.C. § 119(e). The entire disclosure of the xxx Provisional Application is hereby incorporated by reference herein.

TECHNICAL FIELD
[0002] This disclosure relates generally to toys and, specifically, to air blasters or air cannons. More specifically, this disclosure relates to a means to "shoot" air from a collapsible bladder through an opening providing a "blast" of air toward a target.
SUMMARY
[0003] Disclosed are systems, devices, and/or methods of use thereof regarding toys and, specifically, to air blasters or air cannons. In various aspects, an air cannon includes a handle having a handle body and a ring at a proximal end of the handle body. The air cannon may also include a bag extending between a first end and a second end, the first end of the bag securable to an interior of the ring. The air cannon may further include a coil disposed about an interior circumference of the bag, with the coil extending from a first end to a second end. In some embodiments, the first end of the coil is engaged by the interior of the ring to secure the first end of the bag to the interior of the ring. The air cannon may additionally include a face plate, or face pad, connected to a first side of the ring and a back pad permanently, or reversibly, connected to the second end of the bag.
[0004] In another aspect, an air blaster toy includes a handle body and a ring extending from an end of the handle body. The air blaster toy may also include a bag secured to an inside of the ring at a first end of the bag, and a coil disposed about a circumference of the bag. In some embodiments, a first end of the coil is secured to the inside of the ring. The air blaster toy may further include a face plate, or face pad, connected to a first side of the ring and a back pad connected to a second end of the bag, where the second end of the bag is opposite the first end of the bag. The face plate may further define an opening of a smaller diameter than the ring to allow air to pass through.
[0005] Other aspects of the disclosed subject matter, as well as features and advantages of various aspects of the disclosed subject matter, should be apparent to those of ordinary skill in the art through consideration of the ensuing description, the accompanying drawings, and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0006] In the drawings:
[0007] FIG. **1** illustrates a front perspective view of an air cannon or air blaster toy according to the present disclosure;
[0008] FIG. **2** illustrates a back perspective view of the air cannon of FIG. **1**;
[0009] FIG. **3** illustrates a front exploded view of the air cannon of FIG. **1**;
[0010] FIG. **4** illustrates a back exploded view of the air cannon of FIG. **1**;
[0011] FIG. **5** illustrates a front perspective view of the air cannon of FIG. **1** with the face plate removed;
[0012] FIG. **6** illustrates a back perspective view of a handle portion and ring portion of the air cannon of FIG. **1**;
[0013] FIG. **7** illustrates a close-up view of a connection between a bag and a portion of a handle for the air cannon of FIG. **1**;

[0014] FIG. **8**A illustrates a front view of the air cannon of FIG. **1**;
[0015] FIG. **8**B illustrates a compressed configuration side view of the air cannon of FIG. **1**;
[0016] FIG. **9** illustrates a perspective of an alternate embodiment air cannon of FIG. **1**; and
[0017] FIG. **10** illustrates a front perspective view of different size air cannons of FIG. **9**.

DETAILED DESCRIPTION

[0018] Air cannons or air blasters typically allow a user to expel an amount of air or other fluid from within a receptacle. Air blasters can be used to shoot compressed air at a target area. Such compressed are may be “shot” in order to clean the target area. Air blasters can be used to shoot projectiles, such as darts or balls, at targets. Air blasters can also be used to blast or shoot “balls” of air at targets, such as people. In order to create a “blast” of air typically a smaller opening is required to force air through an opening that is smaller in size than the receptable being compressed causing a “cannon” of air forced through the smaller opening.

[0019] Disclosed are systems, devices, and/or methods of use thereof regarding toys and, specifically, to air blasters or air cannons. In various aspects, an air cannon includes a handle having a handle body and a ring, or extension, at a proximal end of the handle body. The air cannon may also include a bag, bladder or receptacle, extending between a first end and a second end, the first end of the bag securable to an interior of the ring. The air cannon may further include a coil, or spring, disposed about an interior circumference of the bag, with the coil extending from a first end to a second end. In some embodiments, the first end of the coil, or the bag itself, is engaged by the interior of the ring to secure the first end of the bag to the interior of the ring. The air cannon may additionally include a face plate connected to a first side of the ring and a back pad permanently, or reversibly, connected to the second end of the bag.

[0020] In another aspect, an air blaster toy includes a handle body and a ring attachable to an end of the handle body. The air blaster toy may also include a bag secured to an inside of the ring at a first end of the bag, and a coil disposed about a circumference of the bag. In some embodiments, a first end of the coil is secured to the inside of the ring. The air blaster toy may further include a face plate, or face pad, connected to a first side of the ring and a back pad connected to a second end of the bag, where the second end of the bag is opposite the first end of the bag.

[0021] FIG. **1** illustrates a front perspective view and FIG. **2** illustrates a back perspective view of an air cannon or air blaster toy **100** according to the present disclosure. Referring to FIGS. **1-2**, as illustrated, the air cannon **100** includes a bag **10**, or receptable, a coil **12**, or spring, a face plate **14**, a back pad **16**, and a handle assembly **18**. The bag **10** may be substantially hollow in order to hold an amount or volume of air within an interior of the bag **10**. The coil **12** may be disposed about a circumference of the bag **10**. For example, in some embodiments, the coil **12** may be disposed about an internal circumference of the bag **10**. In other embodiments, the coil **12** may be disposed about an external circumference of the bag **10**. In some embodiments, the coil **12** may be disposed within a pocket or channel of the bag **10**, where the pocket or channel of the bag **10** can be associated with the internal or external circumference of the bag **10**, wherein the coil **12** may reside entirely within the channel and is not entirely or partially exposed. The channel bag channel **17**, or pocket, may wind around the bag in the same pattern as the coil **12** itself. The face plate **14**, or front plate, may define an outlet **15**, or opening, through which air contained within the bag **10** can be expelled. The opening **15** may be axially aligned with the ring **21** and the bag **10** and may be a smaller diameter than the ring **21**. Alternatively, the opening **15** may not be axially aligned with a center of the ring **21**. Further the opening **15** may include a plurality of openings **15**, one of which may be axially aligned with the center of the ring **21** and other openings **15** that may not be axially aligned.

[0022] The coil **12** and the bag **10** may resemble a tubular member that's substantially circular from a front or back perspective. Alternatively, the coil **12** and bag **10** may be any type of polygonal shape that would allow for collapse and extension of the bag **10** and coil **12**. The bag **10** may include a circumferential wall that creates a void, or space, or hollow portion, within the circumferential wall and extends between a first end **11**, or distal end, of the bag **10** and a second end **13**, or proximal end of the bag **10**. Specifically, the air cannon **100** may have an open or expanded configuration (as illustrated in FIGS. **1-2**) and a closed or compressed configuration (see FIG. **8**B). In use, a user may grasp the handle body **20** and slam their hand or otherwise apply a force against the back pad **16**, or back plate, to compress the bag **10** and the coil **12**. Upon compression of the bag **10** and the coil **12**, any air contained within the bag **10** will be forced out of the outlet **15** within and defined by the face plate **14**. In this way, the air cannon **100** can be used to shoot streams, balls, or puffs of air out of the outlet **15**. In some embodiments, the outlet **15** may be shaped, sized, and configured to create a sound as air is expelled and blasted out of the outlet **15**. In other embodiments, no sound is created as air is expelled.

[0023] The coil **12** provides structural support to the bag **10** as well as a spring function. The bag **10** provides a restrictive support to the coil maintaining the coil **12** around the perimeter of the bag **10** and on a front and back end of the bag. The coil **12** provides a resting open or expanded configuration of the air cannon **100** that is held in its expanded state by the bag **10**. The closed or compressed configuration wherein the coil **12** is compressed that collapses the bag and expels any air within the void or cavity. Because of the nature of the coil it acts as a spring and returns to its resting, expanded position. A use may then easily and repeatedly compress, or constrict, the coil **12** and collapse the bag **10** having it returning to its expanded configuration each time force is no longer applied to the back pad **16**.

[0024] The bag **10** may be constructed from an airtight material, semi-airtight, or partially airtight to ensure no air leaks out of the bag **10** and all, or most of, the air contained within the bag **10** can be expelled from the outlet **15** as the bag **10** is collapsed by an application of force (e.g., slamming, striking, ramming, and other forces). In some embodiments, the bag **10** may be made from nylon, polyester, a woven material, a woven material having a coating to provide an airtight quality, a non-woven material, a non-woven material having a coating to provide an airtight quality, a polymer and/or combinations thereof.

[0025] The handle assembly **18** may include a handle body **20**, or simply a body, and a ring **21**, or extension, attached to the handle body **20** (e.g., at a proximal or top end of the handle body **20**). As seen in FIG. **2**, the bag **10** may be attached or connected to an interior of the ring **21**. Specifically, the bag **10** may extend from the front or first end **11**, or distal end, to the back or second end **13**, or proximal end, where the second end **13** is opposite the first end **11**. Concomitantly, the coil **12** may extend between a front or first end coil ring **11***a* and a back or second end coil ring **13***a*, where the first and second ends **11***a*, **13***a* of the coil **12** substantially correspond to the first and second ends **11**, **13** of the bag **10**. The first end **11** of the bag **10** is received by an interior of the ring **21**. As discussed more fully below with respect to FIG. **7**, in some embodiments, the first end **11** of the bag **10** is irreversibly (i.e., permanently), or reversibly, received by the interior of the ring **21**. Alternatively, the ring **21**, or extension, may be a body itself without a or handle body. A user may hold each end of the extension **21** and utilize the air cannon as well.

[0026] The coil **12** may include the front end coil ring **11***a* and the back end coil ring **13***a* corresponding with the front end **11** of the bag and the back end **13** of the bag. The coil rings may provide structural integrity to the bag **10** to maintain the tubular shape of the bag **10**.

[0027] The back pad **16** may be permanently adhered, coupled to, or otherwise attached to the second end **13** of the bag **10**, or reversibly adhered in a like manner. In some embodiments, the back pad **16** is permanently, or reversibly, attached to the second end **13** of the bag **10** and the second end **13***a* of the coil **12**. In some embodiments, the face plate **14** and the back pad **16** are made from the same material, such as plastic, acrylic, wood, metal, a hard polymer, and/or combinations thereof (e.g., a composite material). Importantly, the material of the face and back pads **14**, **16** needs to be sturdy enough to receive a slamming or positively applied force without buckling, so that the positively applied force can collapse the back end **13** of the bag **10** towards the front end **11** of the bag **10** (and, the back end **13***a* of the coil **12** towards the front end **11***a* of the coil **12**). Additionally, the applied force may collapse the bag **10** itself entirely or partially when the force is applied. In the alternate the ring **21** may be forced toward the back pad **16** in a motion that brings the distal end **11** toward the proximal end **13** or stated otherwise collapse the front end **11** toward the back end **13**.

[0028] FIG. **3** illustrates a front exploded view and FIG. **4** illustrates a back exploded view of the air cannon or blaster **100** of FIG. **1**. Referring to FIGS. **3-4**, as before, the air cannon **100** includes the bag **10**, the coil **12**, the face plate **14**, the back pad **16**, and the handle assembly **18**. The bag **10** extends between the front or first end **11** and the back or second end **13**, with the coil **12** similarly extending between the front end coil ring **11***a* and the back end coil ring **13***a*.

[0029] As illustrated, the handle assembly **18** includes the ring **21** and a handle body **20**. In some embodiments, the handle body **20** includes a first portion **20***a* that is connected or attached to the ring **21**. The handle body **20** may also include a second portion **28** that provides additional structural integrity to the first portion **20***a*. In some embodiments, the second portion **28** is removably connected to the first portion **20***a*, such as through clips, screws, snaps, ties, adhesive, or another appropriate fastening mechanism. In other embodiments, the second portion **28** is irremovably connected to the first portion **20***a*. In some embodiments, the first and second portions **20***a*, **28** may be a single, integrated piece forming the handle body **20**.

[0030] The handle body **20** may further include a decorative plate **26**. In some embodiments, the decorative plate **26** includes indicia, such as a logo, company name, design, decoration, or other graphic. In some embodiments, the decorative plate **26** is a solid color. In other embodiments, the decorative plate **26** is not included. The decorative plate **26** may further provide further structural integrity to the handle to allow for a robust grip for a user to maintain while utilizing the air cannon **100**.

[0031] The back pad **16** may be integrated with or substantially a part of the bag **10** wherein the back pad **16** may reside reversibly or irreversibly within a pocket or sleeve of the back end **13** of the bag **10**.

[0032] FIG. **5** illustrates a front perspective view of the air cannon of FIG. **1** with the face plate **14** removed. The ring **21** may include a first outer portion **22***a* and a second inner portion **22***b*. The first portion **22***a* may surround, encircle, and/or receive the second portion **22***b*. For example, the second portion **22***b* may clip in and/or be adhered to the first portion **22***a*; alternatively, the second portion **22***b* may be held within the first portion **22***a* by a tension or friction fit. In some embodiments, the first and second portions **22***a*, **22***b* are integral and formed as one unitary piece.

[0033] The ring **21** of the handle assembly **18** defines an opening **31** that is in fluid communication with both the interior of the bag **10** and the outlet **15** of the face plate **14** (not illustrated in FIG. **5**). The size and shape of the opening **31** may correspond to a size and shape of the bag **10**. For example, as illustrated, the opening **31** is circular, or substantially circular, and corresponds to the circular configuration of the bag **10**. In some embodiments, a circumference of the opening **31** is determined by a circumference of the second portion **22***b* or the ring **21** itself.

[0034] FIG. **6** illustrates a back perspective view of the handle assembly **18** for the air cannon **100** of FIG. **1**. Specifically, as illustrated, the handle assembly **18** includes the ring **21** attached to the first portion **20***a* of the handle body **20**. As described with respect to FIG. **5**, the ring **21** includes the first half **22***a* and the second half **22***b*, which fit together to form the ring **21** and define the opening **31**. The ring **21** has an outer surface **23***b*, which may correspond to an outer surface of the first portion **22***a*. Disposed on or extending from the outer surface **23***b* may be one or more protrusions **30**, which may facilitate attachment of the air cannon **100** to a backpack, beach bag, bag, purse, etc. to easily transport the air cannon **100**. Additionally, the protrusions **30** may provide for attachment of a cord, robe, yarn or the like between two protrusions **30** to hold the bag **10** of the air cannon **100** in a collapsed, or closed, configuration. The ring **21** also has an inner surface **23***a*, which may correspond to an inner surface of the first portion **22***a* and an inner surface of the second portion **22***b*.

[0035] In some embodiments, the second portion **22***b* includes one or more (e.g., a plurality of) fastening mechanisms for engaging and securing the bag **10** and the coil **12** within the ring **21**. As illustrated, the one or more fastening mechanisms include one or more clips **24**. The one or more clips **24** may include four, five, six, eight or more clips **24** as appropriate to tightly secure the bag **10** and/or the coil **12** to the ring **21**. The front end coil ring **11***a* may slide within the clips **24**. The number of clips **24** utilized to secure the bag **10** and/or the coil **12**, and more specifically the front coil ring **11***a*, may correspond to a size and shape of the clips **24**. In some embodiments, the fit between the first and second portions **22***a*, **22***b* of the ring **21** has enough flex to just barely separate the portions **22***a*, **22***b* in order for a user to insert the bag **10** and/or the coil **12** between them, but is tight enough that the two portions **22***a*, **22***b* cannot be fully separated from each other. In other embodiments, the bag **10** and/or the coil **12**, or front coil ring **11***a*, are attached and secured between the portions **22***a*, **22***b*, and the portions **22***a*, **22***b* cannot be separated.
[0036] FIG. **7** illustrates a close-up view of a connection between a bag **10** and a portion of a handle assembly **18** for the air cannon **100** of FIG. **1**. In some embodiments, the first end **11** of the bag **10** is irreversibly (i.e., permanently), or reversibly, received by the interior of the ring **21**. For example, the first end **11** of the bag **10** may be permanently, or reversibly, received between the first portion **22***a* and the second portion **22***b* of the ring **21**. A ring channel or void **25** may be defined between the first portion **22***a* and the second portion **22***b*, where the ring channel **25** extends around the entire circumference of the first and second portions **22***a*, **22***b*. As clearly visible in FIG. **7**, the coil **12** and the first end **11** of the bag **10** are received within the ring channel **25**. In some embodiments, the clips **24** may include a notch, or barb, or shoulder, or other projection to prevent the coil **12**, or front coil ring **11***a*, and/or the bag **10** from slipping out of the ring channel **25**.
[0037] In some embodiments, a fit between the first portion **22***a*, the clips **24**, and the second portion **22***b* is tight enough to prevent the coil **12** and/or the bag **10** from slipping out of or being dislodged from the ring channel **25**. A friction of the material of the bag **10** may also contribute to preventing the coil **12** and/or the bag **10** from slipping out of or being dislodged from the ring channel **25**. In some embodiments, the fit between the first portion **22***a*, the clips **24**, and the second portion **22***b* is substantially irreversible such that the bag **10** and/or the coil **12** cannot be removed from the ring **21**. An adhesive or additional attachment mechanism may be utilized to further secure the bag **10** and/or the coil **12** between the first portion **22***a* and the second portion **22***b*.
[0038] Alternatively, the clips **24** may allow for the removal of the bag **10** from the ring **21** such that the bag **10** may be replaced with a new bag **10** in the event of damage to the original bag **10**.
[0039] The face plate **14** may be attached, adhered to, or otherwise interface with the ring **21** on a front or first side of the ring **21**. The face plate **14** may be molded as part of the ring **21** (as further described with respect to FIG. **9**, herein). The face plate **14** may correspond to the opening **31** size wherein the face plate **14** may fit complementary within or on the ring **21**. The back or second side of the ring **21** may correspond to the ring channel **25** defined between the first and second portions **22***a*, **22***b* of the ring **21**; the back side of the ring **21** is opposite the first side of the ring **21**. In some embodiments, the face plate **14** attaches to or otherwise interfaces with the first portion **22***a* of the ring. The face plate **14** may be positioned flush with an edge of the ring **21** or may extend outward, opposite the bag.
[0040] FIGS. **8**A-**8**B illustrate various measurements, in centimeters, for the air cannon **100** of FIG. **1**. Specifically, FIG. **8**A illustrates a width of the air cannon **100**, which substantially corresponds to a width or diameter of the ring **21** portion of the handle assembly **18**. The width of the air cannon **100** may range from approximately 155 mm to 175 mm, such as 160, 163, 166, 169, 170, 173 mm, or a width within a range defined by any two of the foregoing values. A length of the air cannon or blaster **100** may range from approximately 280 mm to 320 mm, such as 285, 290, 295, 298, 300, 305, 310 mm, or a length within a range defined by any two of the foregoing values. Even a much larger air cannon of dimensions of 500 mm wide and 1,000 mm long is contemplated. A thickness

of the air cannon or blaster **100**, as shown in FIG. **8**B, in a closed configuration may allow it to collapse in a range from approximately 25 mm to 50 cm, such as 30, 34, 38, 40, 45 mm, or a thickness within a range defined by any two of the foregoing values. While these values may reflect a relative easy handheld version of the device, it will be appreciated that much larger and much smaller variations are contemplated insomuch that the sizes may include a width of the air cannon, from a front perspective, with a diameter of a meter or down to 100 mm.

[0041] FIG. **9** illustrates a front perspective of another embodiment of an air cannon or air blaster **200** according to the present disclosure. As before, the air blaster **200** includes the same or similar features as the previous embodiment with the bag **10** and the handle assembly **18**. The handle assembly **18** includes the ring **21** and the handle body **20** as well as the face plate **14**. The air blaster **100** also includes the coil **12**, which, in this embodiment, has been secured within pockets or bag channel(s) **17** of the bag **10**. As illustrated, the pockets **17** are disposed around an outer surface or outer circumference of the bag **10** coiling around the bag **10** like a spring; however, the pockets **17** may be disposed around an inner surface or inner circumference of the bag **10**. Also as illustrated, the pockets **17** are spiraled around the bag **10** in a sort of corkscrew orientation; however, other configurations or shapes of the pockets **17** are contemplated herein.

[0042] The air blaster **200** in this instance includes each of the previous pieces of the pervious embodiment, the face plate **14** with the air outlet **15**, the ring **21**, and the handle body **20** formed as one piece of the handle assembly **18** are integrated into a unitary member. In some embodiments, such as illustrated, the face plate **14** is continuous and integral with the ring **21** and the handle body **20**. That is, the face plate **14**, the ring **21**, and the handle body **20** are formed as one piece, or single unitary piece. Alternatively, the face plate **14** may be continuous and integral with the ring **21**, and the handle body **20** is separately attached to the face plate **14**/ring **21** assembly. Though not illustrated, the air blaster **200** also includes the back pad **16**, upon which a user may exert a force in order to collapse the bag **10** and expel air out of the outlet **15**. In this embodiment the handle body **20**, ring **21**, and face plate **14** may be manufacture as a single injection-molded piece utilizing a rigid polymer; however, other materials are contemplated herein such as metals, alloys, woods, carbon-fiber and the like.

[0043] As before, the bag **10** of the air cannon or blaster **200** is either permanently or reversibly secured to the interior of the ring **21** (see FIG. **7**) as previously described herein; however in this instance the ring **21** itself solely comprises the clips **24** rather than an integration between two portions of the ring **21**. The ring **21** may comprise an inner wall and an outer wall. The inner wall of the ring **21** may be the wall that engages the bag **10** and coil **12**. In this instance the clips **24** of the ring **21** may be positioned on the inner wall of the ring and engage the first end **11** of the bag **10**, with the first coil ring **11***a*, is irreversibly (i.e., permanently), or reversibly, received by the interior of the ring **21**. For example, the first end **11** of the bag **10** may be permanently, or reversibly, received within the ring channel **25** that may be defined between the clip **24** and the ring **21** where the ring channel **25** extends around the entire circumference of the inner wall, or interior, of the ring **21**. In some embodiments, the clips **24** may include a notch, or barb, or shoulder, or other projection to prevent the coil **12**, or front coil ring **11***a*, and/or the bag **10** from slipping out of the ring channel **25**.

[0044] Alternatively, the bag **10**, with the coil **12**, may reside within the ring **21**, without clips **24**, through a frictional or substantially frictional fit that may require substantial force to remove the bag **10** from the ring **21**. Further, the ring channel **25** may extend within the inner wall, or interior, of the ring **21** without any clips wherein the ring channel **25** is configured to received the front coil ring **11***a* such that the front coil ring **11***a* flexes into the channel providing a reversible, albeit strongly frictional, fit. It will be appreciated that a lip or other means within the ring channel **25** are contemplated to make the bag **10**, with the coil **12**, fit more secure within the ring **21**.

[0045] Similarly, the back pad **16** may be either irreversibly secured to the back or second end **13** of the bag **10** (see FIG. **2**), or the back pad may be reversibly fixed to the back end **13** of the bag

10.
[0046] Referring to FIG. **10** the air cannon **200** is illustrated with different sizes to showcase the ability to utilize the design described herein.
[0047] The method of manufacture, or assembly, for the air cannon **100** may be as described substantially herein with multiple pieces fixed to each other, particularly the face plate **14** to the ring **21** and to the handle **20**, to produce a functional air cannon **100**. Alternatively, the handle **20**, ring **21**, and face plate **14** may be formed of a single unitary piece with attachment to the bag **10**. The bag **10** is contemplated to integrally include the coil **12** and the back pad **16** with a simple fastening mechanism to the handle body **20** through means described herein.
[0048] While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the spirit and scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It should also be noted that some of the embodiments disclosed herein may have been disclosed in relation to a particular air blaster (e.g., an air cannon); however, other fluids (e.g., water, gas, etc.) are also contemplated.
[0049] The terms “a,” “an,” “the,” and similar referents used in the context of describing the embodiments of the present disclosure (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein is intended merely to better illuminate the embodiments of the present disclosure and does not pose a limitation on the scope of the present disclosure. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the embodiments of the present disclosure.
[0050] Groupings of alternative elements or embodiments disclosed herein are not to be construed as limitations. Each group member may be referred to and claimed individually or in any combination with other members of the group or other elements found herein. It is anticipated that one or more members of a group may be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.
[0051] Although this disclosure provides many specifics, these should not be construed as limiting the scope of any of the claims that follow, but merely as providing illustrations of some embodiments of elements and features of the disclosed subject matter. Other embodiments of the disclosed subject matter, and of their elements and features, may be devised which do not depart from the spirit or scope of any of the claims. Features from different embodiments may be employed in combination. Accordingly, the scope of each claim is limited only by its plain language and the legal equivalents thereto.

Claims

1. An air cannon comprising: a handle assembly comprising a handle body and a ring at a proximal end of the handle body; a bag comprising a first end and a second end, the first end of the bag securable to the ring; a coil disposed about a circumference of the bag, the coil extending from the first end of the bag and the second end of the bag, wherein the first end of the coil and the bag engage the ring to secure the first end of the to the ring; a face plate; and a back pad connected to the second end of the bag.

2. The air cannon of claim 1, wherein the handle assembly comprises a single unitary piece.

3. The air cannon of claim 1, wherein the ring comprises the face plate wherein the face plate comprises an opening, the opening smaller in diameter than the ring.

4. The air cannon of claim 1, wherein the ring comprises a plurality of clips for securing the first end of the bag and the first end of the coil within the interior of the ring.

5. The air cannon of claim 1, wherein the ring comprises a circumferential channel extending along the inner wall of the ring.

6. The air cannon of claim 1, wherein the bag is substantially tubular and comprises a void between the first send and the second.

7. The air cannon of claim 1, wherein the bag is made from a substantially airtight material.

8. The coil of claim 1 further comprising: a first coil ring positioned toward the first end and a second coil ring positioned toward the second end, wherein the coil, the first coil ring and second coil ring provide structural support to the bag.

9. The coil of claim 8 wherein the first coil ring reversibly engages the ring.

10. An air blaster toy comprising: a handle assembly comprising; a ring; a handle extending from the ring; and a face plate extending within the ring comprising a first opening smaller in diameter than the ring; a bladder secured to the ring at a first end of the bladder; a coil disposed about a circumference of the bladder, a first end of the coil secured to the ring; and a back pad connected to a second end of the bladder, the second end opposite the first end of the bladder.

11. The air blaster toy of claim 10, wherein the handle assembly comprises a single unitary piece.

12. The air blaster toy of claim 10, wherein the ring comprises an inner wall and an outer wall, the inner wall comprising a plurality of clips for securing the first end of the bag and the first end of the coil to the ring.

13. The air blaster toy of claim 10, wherein the plurality of clips reversibly engage the bladder and coil.

14. The air blaster toy of claim 10, wherein the bladder is substantially tubular and comprises a cavity between the first end and the second end.

15. The coil of claim 10 further comprising: a first coil ring positioned toward the first end and a second coil ring positioned toward the second end, wherein the coil, the first coil ring and second coil ring provide structural support to the bladder.

16. The air cannon of claim 10, wherein the bag is made from a substantially airtight material.

17. An air cannon comprising: a handle assembly comprising; a ring; a handle extending from the ring; and a face plate extending within the ring comprising a first opening smaller in diameter than the ring; a collapsible bladder secured to the ring at a first end of the bladder; a coil disposed about a circumference of the bladder within a bladder channel, the coil providing structural support to the bladder and the bladder providing restrictive support to the coil; wherein a first end of the coil is secured to the ring at the same position as the first end of the bladder.

18. The air cannon of claim 17 comprising a first configuration wherein the bladder and coil are expanded.

19. The air cannon of claim 18 comprising a second configuration wherein the bladder is collapsed and the coil is compressed.